Practice workbook · Explain systolic, diastolic and changing pressure around the circulation

Meaning of systolic and diastolic pressure and reading or explaining pressure changes from heart to arteries, capillaries and veins.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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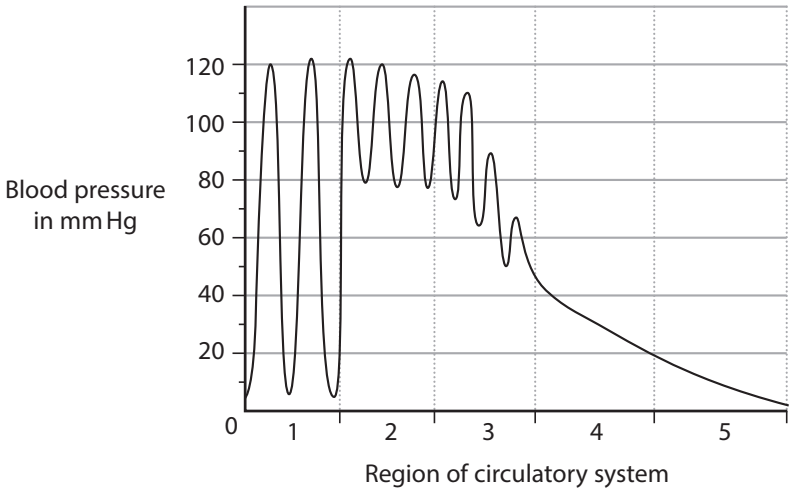
Type ID: HBI-EXPLAIN-BLOOD-PRESSURE-CHANGES

1. Question 3 · part b · 5 marks

June 2023 Paper 1R

Worked-example retry · Attempt part b only.

3 The graph shows how blood pressure varies in different regions of the circulatory system.



(a) (i) Estimate the difference between the maximum blood pressure and minimum blood pressure in region 1.

(2)

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difference in blood pressure = ... mm Hg

(ii) Which part of the circulatory system is represented by region 2?

(1)

- ☒ **A** arteries
- ☒ **B** capillaries
- ☒ **C** heart
- ☒ **D** veins

(iii) Which part of the circulatory system is represented by region 5?

(1)

- ☒ **A** arteries
- ☒ **B** capillaries
- ☒ **C** heart
- ☒ **D** veins

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1. Question 3 · part b · 5 marks

June 2023 Paper 1R

Worked-example retry · Attempt part b only.

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(b) Explain why blood pressure varies in region 1.

(5)

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(c) Explain why the blood pressure changes in region 4.

(4)

(Total for Question 3 = 13 marks)

1. Question 3 · part b · 5 marks · Working and answer

Working and answer

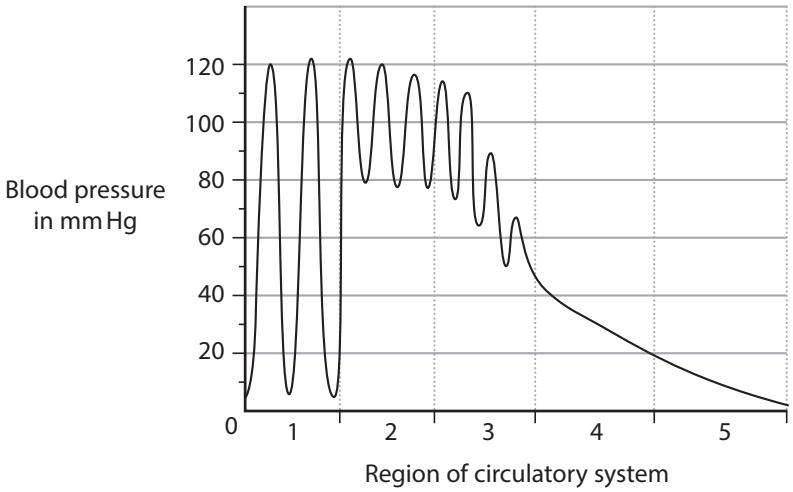
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2. Question 3 · part a.ii · 1 mark

June 2023 Paper 1R

Further practice · Attempt part a.ii only.

3 The graph shows how blood pressure varies in different regions of the circulatory system.



(a) (i) Estimate the difference between the maximum blood pressure and minimum blood pressure in region 1.

(2)

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difference in blood pressure = ... mm Hg

(ii) Which part of the circulatory system is represented by region 2?

(1)

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- ☒ **B** capillaries
- ☒ **C** heart
- ☒ **D** veins

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(1)

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- ☒ **C** heart
- ☒ **D** veins

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2. Question 3 · part a.ii · 1 mark · Working and answer

Working and answer

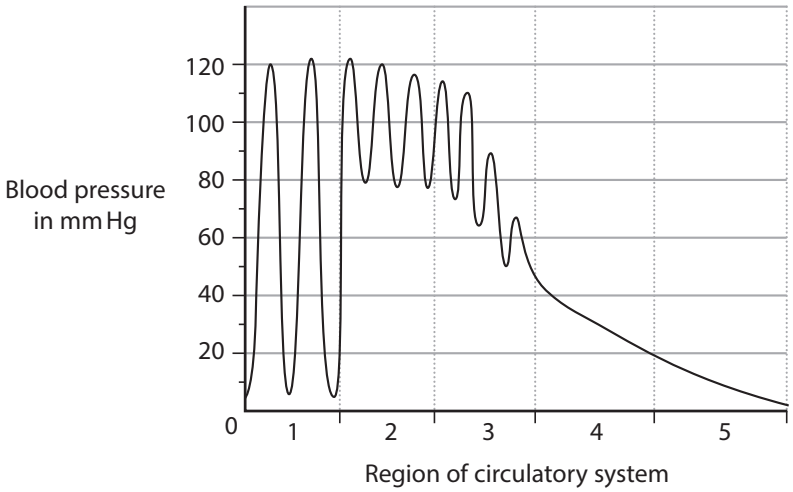
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3. Question 3 · part a.iii · 1 mark

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Further practice · Attempt part a.iii only.

3 The graph shows how blood pressure varies in different regions of the circulatory system.



(a) (i) Estimate the difference between the maximum blood pressure and minimum blood pressure in region 1. **(2)**

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difference in blood pressure = ... mm Hg

(ii) Which part of the circulatory system is represented by region 2? **(1)**

- ☒ **A** arteries
- ☒ **B** capillaries
- ☒ **C** heart
- ☒ **D** veins

(iii) Which part of the circulatory system is represented by region 5? **(1)**

- ☒ **A** arteries
- ☒ **B** capillaries
- ☒ **C** heart
- ☒ **D** veins

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P 7 2 5 8 6 A 0 6 2 0

3. Question 3 · part a.iii · 1 mark

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Further practice · Attempt part a.iii only.

(b) Explain why blood pressure varies in region 1.

(5)

(c) Explain why the blood pressure changes in region 4.

(4)

(Total for Question 3 = 13 marks)

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3. Question 3 · part a.iii · 1 mark · Working and answer

Working and answer

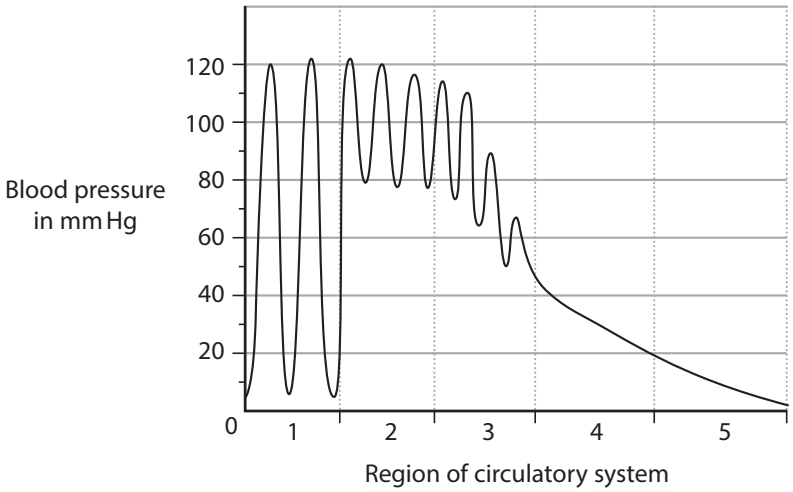
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4. Question 3 · part c · 4 marks

June 2023 Paper 1R

Further practice · Attempt part c only.

3 The graph shows how blood pressure varies in different regions of the circulatory system.



(a) (i) Estimate the difference between the maximum blood pressure and minimum blood pressure in region 1.

(2)

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difference in blood pressure = ... mm Hg

(ii) Which part of the circulatory system is represented by region 2?

(1)

- ☒ **A** arteries
- ☒ **B** capillaries
- ☒ **C** heart
- ☒ **D** veins

(iii) Which part of the circulatory system is represented by region 5?

(1)

- ☒ **A** arteries
- ☒ **B** capillaries
- ☒ **C** heart
- ☒ **D** veins

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4. Question 3 · part c · 4 marks

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Further practice · Attempt part c only.

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(b) Explain why blood pressure varies in region 1.

(5)

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(c) Explain why the blood pressure changes in region 4.

(4)

(Total for Question 3 = 13 marks)

4. Question 3 · part c · 4 marks · Working and answer

Working and answer

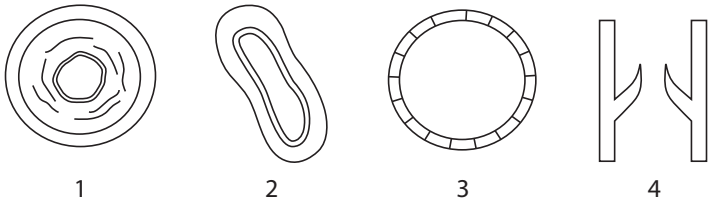
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5. Question 6 · part b · 4 marks

November 2024 Paper 2

Further practice · Attempt part b only.

6 (a) The diagram shows different sections through various blood vessels.



Not to scale

(i) Identify each type of blood vessel.

(4)

blood vessel 1

blood vessel 2

blood vessel 3

blood vessel 4

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5. Question 6 · part b · 4 marks

November 2024 Paper 2

Further practice · Attempt part b only.

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(ii) State one reason for your choice for each blood vessel.

(4)

blood vessel 1

blood vessel 2

blood vessel 3

blood vessel 4

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P 7 5 9 5 9 A 0 1 9 2 4

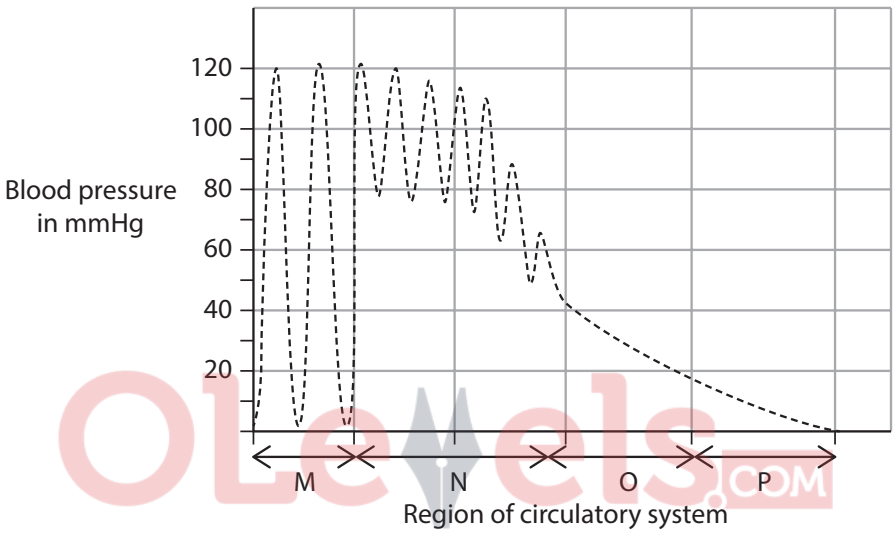
Turn over ▶

5. Question 6 · part b · 4 marks

November 2024 Paper 2

Further practice · Attempt part b only.

(b) The graph shows changes in blood pressure in four regions of the circulatory system labelled M, N, O and P.



Identify each region, M, N, O and P, of the circulatory system.

(4)

M .....

N .....

O .....

P .....

(Total for Question 6 = 12 marks)

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5. Question 6 · part b · 4 marks · Working and answer

Working and answer

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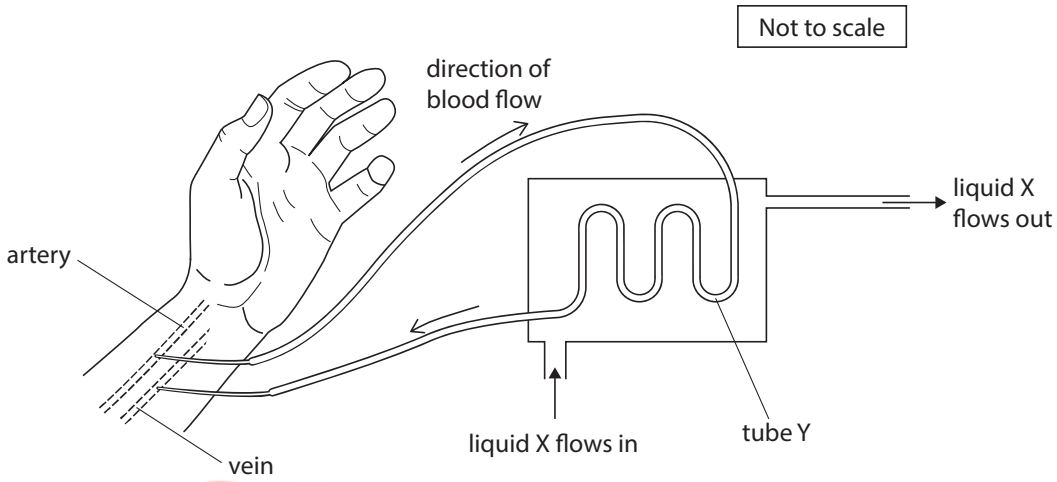
6. Question 5 · part a.i · 4 marks

June 2025 Paper 1R

Further practice · Attempt part a.i only.

5 The diagram shows a kidney dialysis machine.

The wall of tube Y is partially permeable so it allows the passage of small molecules.



(a) (i) Explain why the blood flow is in the direction shown in the diagram.

(4)

(ii) State which part of the kidney nephron has the same role as tube Y.

(1)

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P 7 4 6 2 8 A 0 1 2 2 0

6. Question 5 · part a.i · 4 marks · Working and answer

Working and answer

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7. Question 9 · part b.i · 2 marks

June 2022 Paper 1

Further practice · Attempt part b.i only.

9 In England, 31% of men and 26% of women have high blood pressure.

(a) Explain how medication can be used to treat people with high blood pressure.

(4)

(b) (i) Describe what is meant by systolic blood pressure.

(2)

(ii) Describe what is meant by diastolic blood pressure.

(2)

(Total for Question 9 = 8 marks)

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7. Question 9 · part b.i · 2 marks · Working and answer

Working and answer

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8. Question 9 · part b.ii · 2 marks

June 2022 Paper 1

Further practice · Attempt part b.ii only.

9 In England, 31% of men and 26% of women have high blood pressure.

(a) Explain how medication can be used to treat people with high blood pressure.

(4)

(b) (i) Describe what is meant by systolic blood pressure.

(2)

(ii) Describe what is meant by diastolic blood pressure.

(2)

(Total for Question 9 = 8 marks)

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8. Question 9 · part b.ii · 2 marks · Working and answer

Working and answer

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